Course Overview

Syllabus Review

Course Information

- **Academic Term:** Daytona Beach Spring 2026
- **Term Dates:** January 7, 2026 - April 30, 2026
- **Credit Hours:** 3
- **Mode of Delivery:** In Person
- **Meeting(s):** 11:00 AM - 11:50 AM Monday,Wednesday,Friday
- **Location(s):** Lehman Building 367

Office Hours

- In-Person:
  - Monday, Wednesday, and Friday from 12:30-1:30pm in LB 346
- Virtual:
  - Book a meeting for a time that we both have available using Microsoft Booking.
  - URL: https://outlook.office.com/bookwithme/user/968bb865d5e84a2fa1ec48f65352cdb2@erau.edu

Course Description

Computer engineering aspects; embedded sensor and actuator controllers; high-performance computing systems for air traffic control and weather systems; critical design factors including technological, economic, social, and professional considerations; techniques and practices in database management, human-computer interaction, and system networks.

Course Goals

Students will learn to critically analyze how aviation regulations influence system design and evaluate the effectiveness of various navigation and communication technologies. They will develop the ability to model representative aviation systems using tools such as simulations or hardware to translate technical concepts into clear, actionable designs that address real-world aerospace challenges.

Student Learning Outcomes

1. Describe the breadth of computing systems and practices used in aerospace and aviation, including their technological and engineering components.
2. Analyze the performance of aerospace/aviation computing systems.
3. Evaluate the historical perspectives on computing in aerospace and aviation and their impact on current aviation systems.
4. Engage in debates/discussions about critical computing topics such as digital avionics, flight control systems, and air traffic management.
5. Design fault-tolerant computing systems, flight management systems, or human-computer interfaces (e.g., Glass cockpits, HUDs) using principles learned from case studies.
6. Implement models of avionics systems in simulation or hardware.

Reading Materials

- Most reading material will be made available.
- Optional Textbook:
  - Title: Aircraft Systems
  - Authors: Chis Binns
  - Publisher: IEEE Press
  - Publication Date: 2018
  - Link: https://learning.oreilly.com/library/view/aircraft-systems/9781119259541/
  - Note:
    - The textbook is free under ERAU's license with O'Reilly books. When prompted to log in, use your ERAU email address. Then, select the option for SSO Sign-In. From there, you will sign on through the ERAU sign-in page.
    - This service provides access to a wealth of online textbooks and programming guides

Assessment Activities and Grading

- **Grade Distribution**
  - Participation: 20%
  - Homework: 20%
  - Project: 20%
  - Exam 1: 20%
  - Exam 2: 20%
- **Participation** grades shall be assessed through participation in individual and group in-class activities including pop quizzes.
- **Homework** shall be assigned for each module.
- **Exam** grades shall be assessed by two exams, a midterm through module 5 and a non-comprehensive final exam, exam 2. Exam 2 is scheduled during the final timeslot, Saturday, April 25, 2026 12:30pm-2:30pm.
- **Project** grades shall be assessed through preliminary project milestones, a final presentation, and a report. Module 10 is set aside for relevant tutorials, time to work on your project in-class and final presentations.
- NOTE: Attendance is not explicitly graded; however, if you miss class you will potentially/likely miss an in-class activity that counts toward your participation grade.

AI Policy

- In addition to the University's basic policy as shown in the syllabus, the additional statement applies to this course.

For CEC 300, please feel free to use ChatGPT for studying purposes (e.g. exploring topics, reviewing materials, etc.). However, all written content that you submit for homework, exams, and presentations should be compiled, organized, and written by you. Use of external sources must also be cited. See honor code violation policy (below).

Policies

- **Late Policy:** Assignments are expected to be submitted on time. An 8-hour grace period is permitted for all homework or project deliverables without penalty. From 8 hours-48 hours, you receive a 2.5 percent deduction for every hour that it remains late. After 48 hours, the assignment will be awarded a zero (0). Exceptions include exam absences and presentation absences.
- **Exam Absence:** An unexcused absence to an exam will result in a zero. Under rare circumstances, Dr. Stansbury has the discretion to excuse an exam absence under a **significant** hardship with documentation.
  - ***IMPORANT NOTE:*** personal travel is not and will not be an excuse for missing an exam. The final exam (exam 2) is scheduled for April 25 from 12:30-2:30pm.

Policies (continued)

- **Presentation Absence:** Once you / your group has signed up for a presentation time slot, you are expected to present at that time. Only under significant hardship will a makeup opportunity be granted. Otherwise, a zero is issued.
- **Changes to Course Schedule or Module Topics:** The following schedule represents the tentative schedule for which we will complete this course. The dates and order of materials are subject to change, but notice will be given via Canvas Announcement when such changes are to be made.
  - KEEP IN MIND: this is the second time Dr. Stansbury has taught this course and the program outcomes have changed for this term. Entire modules are being created or rebuilt. Some grace with regard to schedule adjustments is requested and appreciated.

Policies (continued)

- **Honor Code Violations:** All students are expected to follow the ERAU honor code and the expectations laid out above regarding the use of AI/ML technology usage. Dr. Stansbury has established the following guidelines regarding honor code violations. All cases shall follow the University's due process procedures as laid out in the student handbook.
  - First-time, minor infractions can result in a warning and/or loss of points for all or part of the assignment.
  - Multiple and/or more server infractions could result in a zero for an assignment, a drop in course letter grade, or failure for the course based upon the circumstances of the violation.

- **SAS Testing Center and Exams:** If you receive accommodations on your exam by Student Accessibility Services, you must submit your request at least one week prior to the exam.

Date	Day	Module Code	Module Name	Topic
2026-01-07	Wed	M1	Foundations & Historical Evolution of Aerospace/Aviation Computing	Course Introduction; Overview of Aerospace & Aviation Computing
2026-01-09	Fri			Historical Milestones: Apollo, Early Avionics, Spacecraft Computers
2026-01-12	Mon			Evolution of Digital Avionics; Regulatory Landscape
2026-01-14	Wed			Flight Computers: From Analog to Digital
2026-01-16	Fri	M2	Fundamentals of Navigation, Guidance & Control Computing (GN&C)	Aerospace Navigation Sensors (IMU, GPS, star trackers)
2026-01-19	Mon			No Class – MLK Day
2026-01-21	Wed			INS & GN&C Computing Pipelines
2026-01-23	Fri			Introduction to Kalman Filtering (Conceptual)
2026-01-26	Mon	M3	Digital Avionics Architectures & Data Networks	Federated vs IMA Architectures
2026-01-28	Wed			ARINC Standards (429/653/664)
2026-01-30	Fri			Avionics Networking & Latency
2026-02-02	Mon	M4	Flight Control Computing (Aircraft & Spacecraft)	Fly-By-Wire Computing
2026-02-04	Wed			Stability Augmentation & Control Laws
2026-02-06	Fri			Spacecraft Attitude Control Computing
2026-02-09	Mon			Case Studies: Airbus A320, F-16, Falcon 9 Control

2026-02-11	Wed	M5	Flight Management Systems & Human–Computer Interfaces	Flight Management System (FMS) Architecture
2026-02-13	Fri			LNAV/VNAV & Guidance Computations
2026-02-16	Mon			Autopilot Logic & Automation
2026-02-18	Wed			Cockpit Interfaces: Glass Cockpits, HUD, HMI
2026-02-20	Fri			Guest Speaker
2026-02-23	Mon		Exam 1 Review & Exam 1	Exam 1 Review
2026-02-25	Wed			Exam 1
2026-02-27	Fri	M6	Air Traffic Management & Networked Aviation Computing	ATC System Architecture (ERAM, STARS, SWIM)
2026-03-02	Mon			ADS-B, CPDLC, Satellite ATC
2026-03-04	Wed			UAS/UAM Traffic Management Computing
2026-03-06	Fri	M7	Aerospace Computing Beyond Aviation (Spacecraft, Deep Space, Launch)	Spacecraft Onboard Computers
2026-03-09	Mon			Deep Space Navigation & Communications
2026-03-11	Wed			Launch Vehicle Computing
2026-03-13	Fri			Autonomous Docking & Rendezvous Systems
2026-03-16	Mon		Spring Break	Spring Break – No Class
2026-03-18	Wed			Spring Break – No Class
2026-03-20	Fri			Spring Break – No Class

2026-03-23	Mon	M8	Reliability, Fault Tolerance, Safety Analysis	Fault-Tolerant Design Principles
2026-03-25	Wed			Redundancy: Triplex, Quadruplex, Byzantine Tolerance
2026-03-27	Fri			Failure Case Studies (Ariane 5, 737 MAX, Mars missions)
2026-03-30	Mon			Safety Assessment: FMEA, FTA, FHA
2026-04-01	Wed	M9	Security & Assurance in Aerospace Systems	Cybersecurity in Avionics & ATC
2026-04-03	Fri			Securing Satellite & GNSS Systems
2026-04-06	Mon	M10	Capstone Implementation Module (Simulation or Hardware)	Avionics Simulation Lab (in-class)
2026-04-08	Wed			Project Implementation Workshop
2026-04-10	Fri			Guided Project Workday
2026-04-13	Mon			Guided Project Workday
2026-04-15	Wed			Guided Project Workday
2026-04-17	Fri			Final Project Presentations (Groups 1–3)
2026-04-20	Mon			Final Project Presentations (Groups 4–6)
2026-04-22	Wed		Review	Exam 2 Review & Course Synthesis
4/25/2026	SAT		Exam 2 (Final) - Saturday, April 25, 2026 12:30pm-2:30pm	

Questions?

Computing in Aviation and Aerospace
Foundations / Historical Evolution
3
Fundamentals of Guidance Navigation and Control
5
Digital Avionics Architectures / Data Networks
7
Flight Control Computing
7
Flight Management Systems & Human-Computer Interfaces
4
Air Traffic Management & Networked Aviation Computing
8
Aerospace Computing Beyond Aviation
4
Reliability, fault tolerance, and safety
8
Security & Assurance
2

Module 1

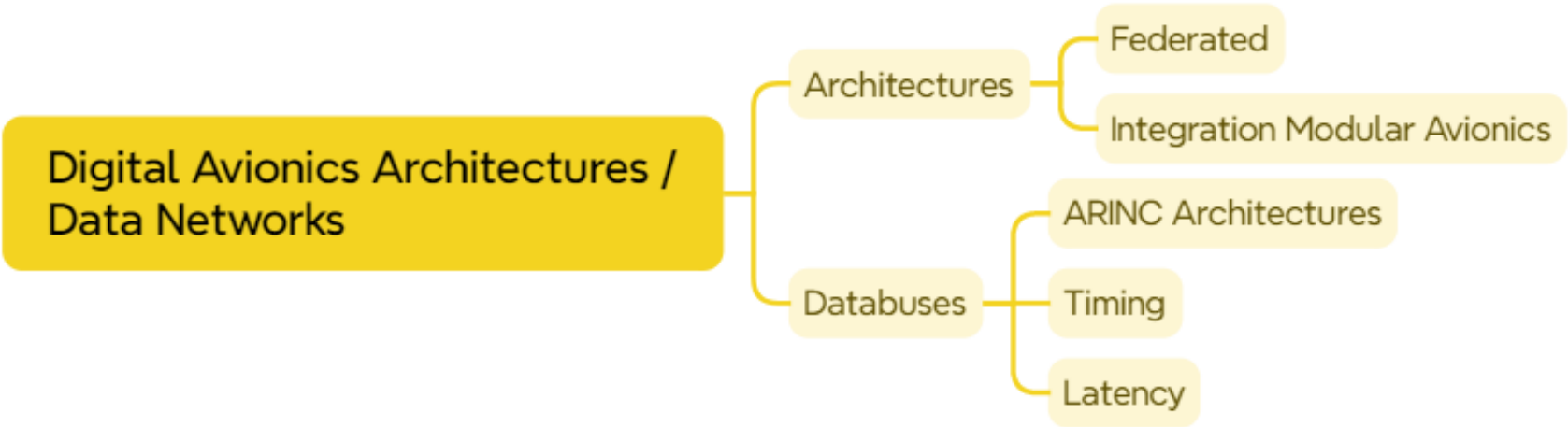
- Establishes a foundation for where we are today.
- The history of computing for aviation and aerospace are intertwined.
- Standards define the requirements for aircraft or spacecraft components, systems, and systems of systems.
- System certification must be addressed for airworthiness or “spaceworthiness” following processes established by a governing agency. In the United States, this is the Federal Aviation Administration.

Module 2

- Sensors provide the necessary inputs to enable guidance, navigation, and control of aviation and aerospace systems (e.g. Inertial Navigation Systems (INS), gyroscopes, magnetometers, compasses, air pressure systems, etc.)

- Guidance Navigation (GNC) and INS must work together to maintain a control loop with sensor inputs being translated to vehicle actuation.

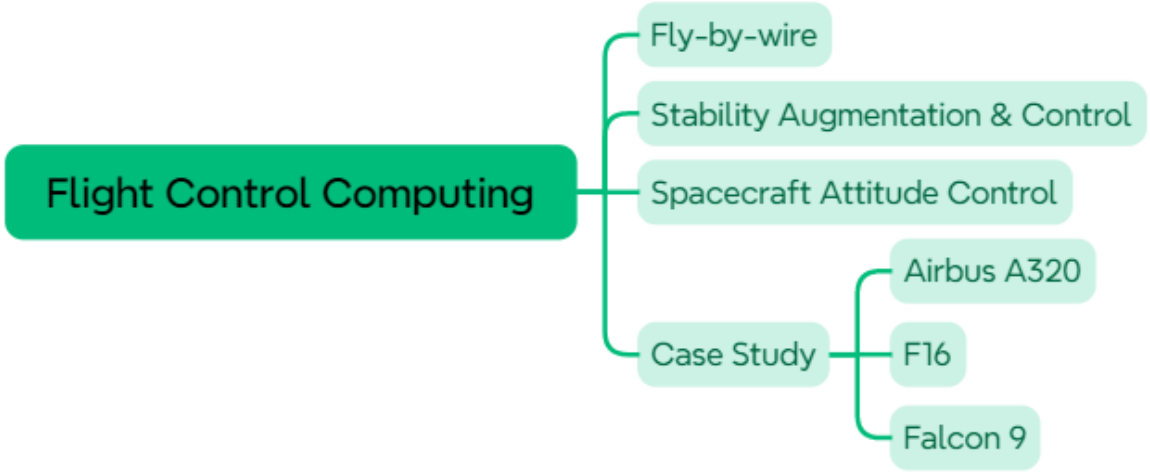
- Signal processing techniques like Kalman Filters provides a means of estimating vehicle state based upon multiple sensory inputs.



Module 3

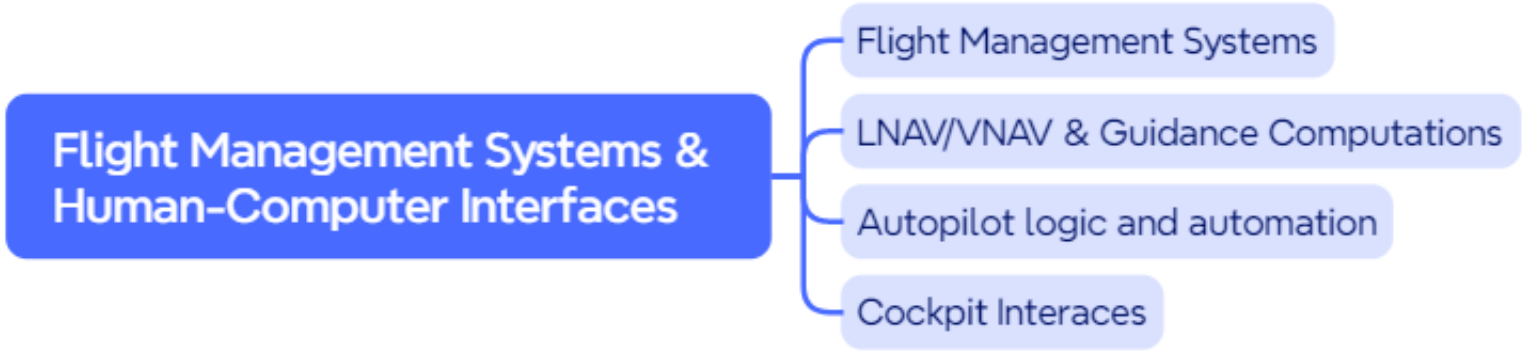
- Avionics architectures can be classified as:
  - Federated (each function is its own line replaceable unit)
  - Distributed (integrated network connecting processors working on shared computing tasks)
  - Integration Modular Avionics (multiple functions integrated into common hardware)

- Databuses enable communication and transfer of data between avionic systems.
  - ARINC is one of several agencies that produce databus standards used in aviation.
  - Timing and latency of a databus architecture must be approximated and improved to meet performance requirements of the air or space vehicle.



Module 4

- Flight control has evolved to eliminate the risk of mechanical failures, reduce size, weight, and power known as **Fly-by-Wire**
  - **Fly-by-Wireless** uses similar concepts with wireless communication
- Control systems enable **flight stability and control** of the vehicle.
- **Spacecraft attitude control** will be examined in this topic.
- **Case studies** of flight control computing will be examined investigating the flight control systems of the Airbus A320, the F-16 fighter jet, and Falcon 9 rocket.

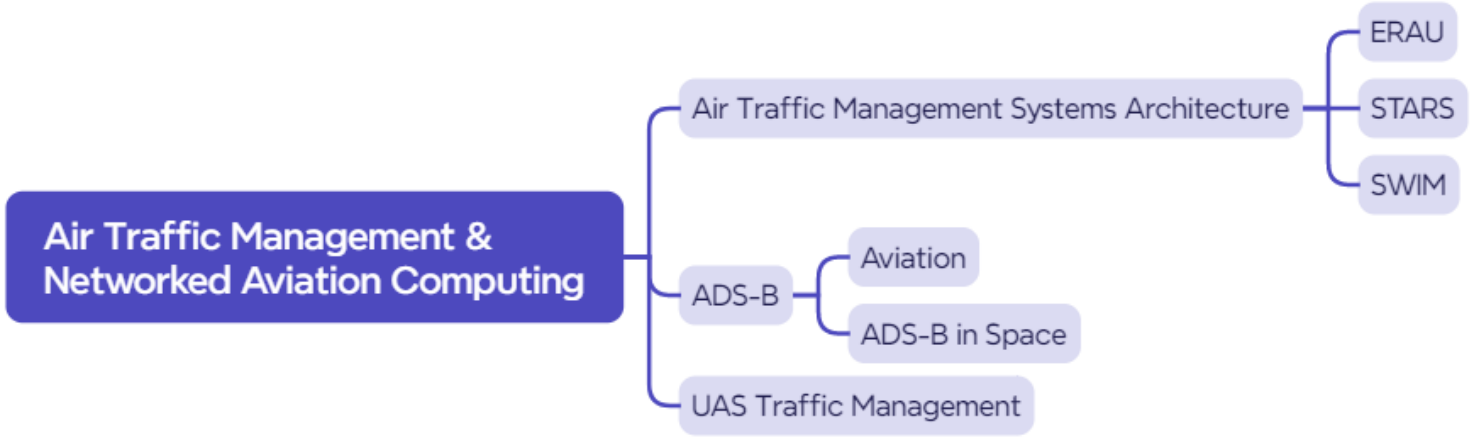


Module 5

- Flight management systems are onboard computers that support flight operations and reduce pilot workload.
  - Management of flight plans
  - Lateral Navigation (LNAV) and Vertical Navigation (VNAV) Guidance

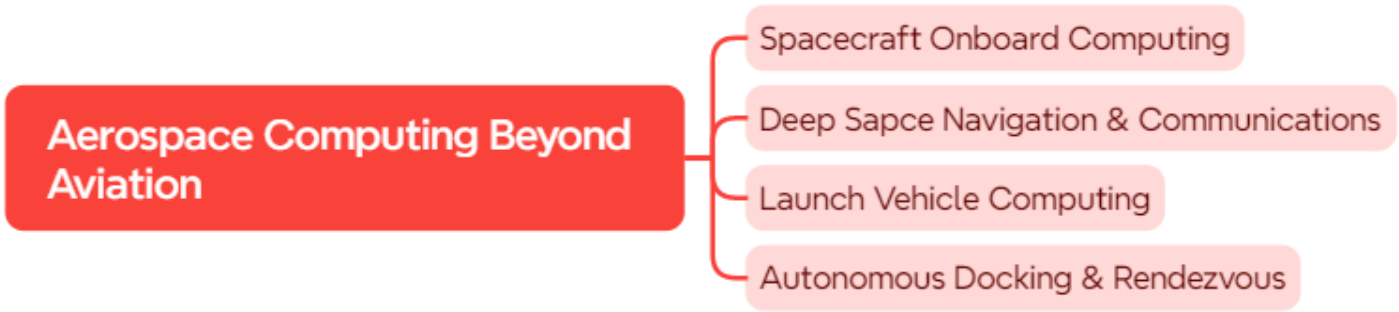
- FMS integrates inputs from GPS, INS, and radio navigation sytems.

- FMS interfaces with with
  - Cockpit interfaces – provide situational awareness and guidance to pilot
  - Autopilot – enables efficiency, safety, and precision of part of or entire flight.



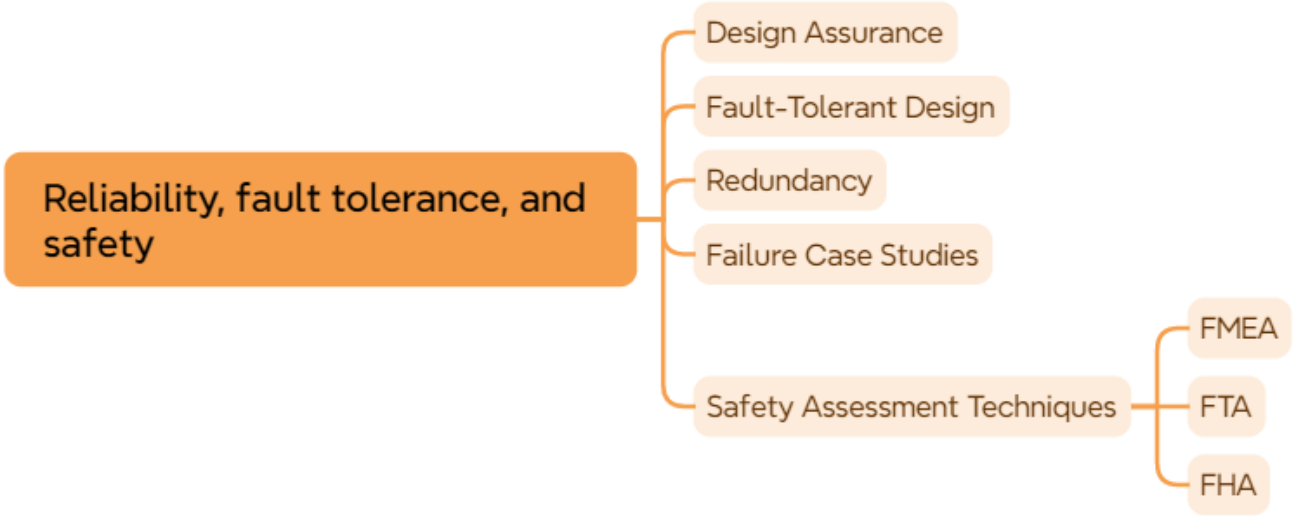
Module 6

- Computing in aviation and aerospace is not limited to vehicle-based systems.
- Air Traffic Management requires significant infrastructure (primarily ground-based, but some space-based systems)
  - Includes:
    - ERAM – enroute airspace management
    - STARS – standard terminal automation replacement system
    - SWIM – system wide information management (infrastructure and standards for information to be exchanged across FAA systems)
- Automatic Dependent Broadcast – Surveillance – tracks aircraft through a network of ground-based terminals.
- UAS Traffic Management and AAM/UAM Traffic Management strategies require significant interconnectivity including a SWIM-like network backbone and vehicle-to-vehicle communications.



Module 7

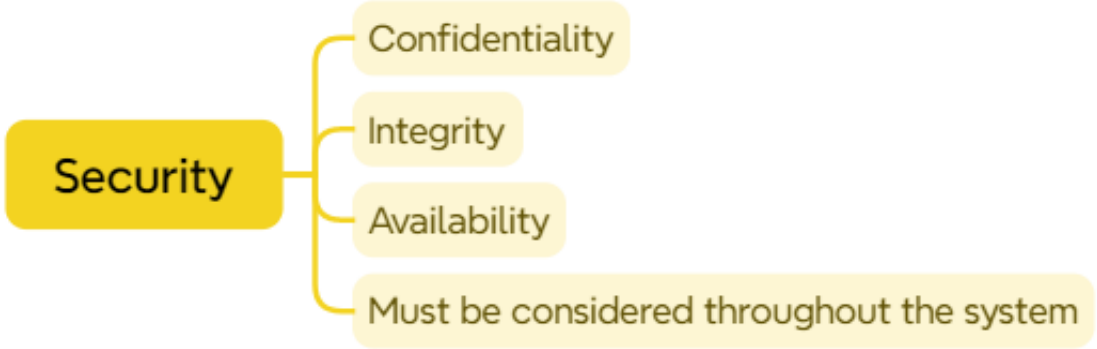
- Module explores topics for space-based systems including:
  - Spacecraft Onboard Computing: architecture, hardware, and software.
  - Deep Space Navigation and Communications are substantially different that aviation-based strategies
    - No compass or GPS to guide the system
    - Latency of communications increases proportionally with distance
  - Launch vehicles require their own computing systems to automate launch functions, monitor vehicle health, maintain desired trajectory, etc.
  - Autonomous docking & rendezvous.
    - In space, automation often guides a space craft's rendezvous with satellites, space stations, etc. and in some cases dock with those systems.
    - Automated return and landing on ground-based or sea-based platforms



Module 8

This module addresses more systems engineering focused topics applicable.

- Design Assurance Levels are a means of identifying the criticality of hardware or software systems under development.
- Standards define the development practice, documentation, verification, and validation of onboard systems.
- Robustness is achieved through **fault-tolerant design and redundancy.**
- Case studies shall be explored for failure cases in aviation and aerospace.
- Safety assessment techniques are necessary for ensuring risk is sufficiently mitigated.
    - Common approaches include:
        - Fault Tree Analysis
        - Failure Modes and Effects Analysis
        - Functional Hazard Assessment



Module 9

- Cybersecurity has become a major consideration for air and space-born systems.
  - Objective to preserve confidentiality, integration, and availability.
- Must be considered across systems and systems of systems:
  - Flight hardware and software,
  - Communications hardware, software, and protocols, and
  - Supporting infrastructure.

Summary

- Course syllabus including policies are reviewed
- Each module addresses one or multiple learning outcomes (see schedule for mapping)
- Computing in Aviation and Aerospace encompasses a variety of systems:
  - Airborne systems (e.g. aircraft), Space-based systems (e.g. spacecraft), and Enabling systems (e.g. ATM, Deep Space Network, etc.)
- The course is under continual improvement such that some modules will be completed as the term progresses.

Questions?